Ahmed Galal Elzeky

AI Engineer | LLMs | RAG | Agentic Systems | Arabic NLP
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Egyptian national | military service: exempted

Microsoft Certified Azure AI Engineer with a published research paper on Arabic LLM reasoning (Elsevier). Builds enterprise generative AI that runs in production: RAG, AI agents and LLM evaluation on a microservice architecture.

Work Experience

AI Engineer **Feb 2025 – Present**
Intelligent Systems (ISYS), IBM partner *Smart Village, Egypt*

- **Built SIA, an enterprise AI platform** for a 1,000-user rollout: FastAPI microservices and Next.js, SSO, MFA, per-customer data isolation and encryption, GDPR compliance, and English/Arabic with Arabic-dialect speech.
- **Built a pluggable provider architecture**: each service runs on the provider the customer licensed in SaaS or installed on-premises, from IBM watsonx, Claude or OpenAI for LLMs to Elasticsearch, Milvus or pgvector for vector search.
- **Built a task-aware model router** that picks the LLM for each task by measured output quality and cost.
- **Raised answer-confidence scoring from AUROC 0.35 to 0.994** and made every answer cite its source document.
- **Built department-gated AI agent tools** (email, calendar, Word and Excel authoring, document comparison, web search) and an **AI recruitment pipeline** for CV parsing, candidate matching and deduplication.
- **Trained Vodafone's engineering team** on IBM watsonx.ai in a 9-hour course (LoRA/QLoRA, prompt tuning, RAG, evaluation, prompt engineering, synthetic data), and pitched ISYS solutions to **C-suite audiences at ISYS conferences**.

Freelance AI & Full-Stack Engineer
Selected client and product work, 2025–2026 *Remote*

- **Built Comic Spot** *(2026)*, a live e-commerce store at comicspot.co, on Next.js and Cloudflare Workers with headless Shopify, where checkout validates every item against Shopify so outdated catalogue data can never complete a purchase.
- **Built Lango** *(2025–2026)*, an offline-first language-learning app, and **designed its core algorithm**: it tracks every word a learner knows and has an LLM write stories of 90% known and 10% new words, then verifies the ratio.

- **AI Developer Intern**, Environ Adapt *(Aug 2024 – Oct 2024)*. Built a Siamese neural network for image similarity that replaced manual bale tagging, plus an image quality-control model.
- **AI Developer Intern**, ESLSCA University *(Feb 2024 – Apr 2024)*. Built a GPT-based admissions assistant that answered applicants and checked their documents.
- **Software Developer Intern**, IIG-HEC *(Jun 2023 – Dec 2023)*. Built the company website and a real-time blog publishing system on React and Firebase.

Publications & Awards

- **Published** "Multi-Hop Arabic LLM Reasoning in Complex QA", *Procedia Computer Science* 244 (2024) 66–75, Elsevier; third of five authors. Presented at ACLing 2024 (Dubai); ESLSCA research award.
- **Winner, GenAI Hackathon** hosted by Stakpak. Built in 48 hours the prototype that became Lango.
- **Times Higher Education Awards Arab World 2024**, Technological/Digital Innovation of the Year, university AI team.
- **Top-3 finalist, GESAwards Africa 2024** representing Egypt and North Africa with GPTube, a graduation project turning YouTube videos into quizzes and summaries with LangChain and OpenAI.
- **Tech content creator** with 37K followers, 484K likes and a 917K-view video, building an Arabic-speaking tech audience.

Skills

AI & LLMs: RAG with hybrid search and reranking (Elasticsearch, Milvus, pgvector), AI agents and multi-agent orchestration, LangGraph, LangChain, MCP, function calling, structured outputs, guardrails

LLM Ops: LLM-as-judge and RAGAS evaluation, Langfuse observability, LoRA/QLoRA fine-tuning, quantisation, self-hosted model serving, prompt engineering

Machine Learning: supervised and unsupervised learning, deep learning, NLP, computer vision, PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn

Programming: Python, TypeScript/JavaScript, C#, SQL

Backend & Frontend: FastAPI, Django, NestJS, microservices, PostgreSQL, MongoDB, Redis, React, Next.js, React Native

Cloud & DevOps: Azure, IBM watsonx, Google Cloud, Docker, Jenkins CI/CD, Cloudflare Workers, Supabase, Git

Certifications

Microsoft Certified: Azure AI Engineer Associate (earned May 2025). Credential ID FA9AAA1E4EAF1DC

IBM: watsonx.ai and watsonx Orchestrate Practitioner Advanced (2025), plus 22 further IBM badges

Education

B.Sc. Computer Science: ESLSCA University, Cairo. GPA 3.5/4.0. Sep 2021 – Sep 2024

Languages: English fluent (IELTS 7.5). German A1 (learning). Japanese JLPT N4. Arabic native